

# Nevada County Experience

Visitor planning platform for **Western Nevada County** — feature overview & demonstration script

## What this is

A purpose-built tourism platform for **Western Nevada County** — the Gold Country foothills region centered on Nevada City and Grass Valley, with full coverage of Penn Valley, North San Juan, Rough & Ready, Washington, Chicago Park, Smartsville, and adjacent Colfax. (Truckee and the Sierra-side areas of Nevada County are out of scope for this build.)

The platform combines **161 curated experiences** with **460+ live-scraped events** from KVMR, Eventbrite, the Nevada City Chamber, the Grass Valley Chamber, and other local sources. Visitors discover what to do; the chamber team curates and approves what they see; the system runs without ongoing developer involvement.

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Geographic scope       | Western Nevada County (Gold Country foothills) + adjacent Colfax                                          |
| Total experiences      | 161 across 11 communities                                                                                 |
| Live event sources     | 6 — KVMR, Eventbrite, NC Chamber, GV Chamber, The Union, Go Nevada                                        |
| Event count (approved) | 460+                                                                                                      |
| Discovery vibes        | 9 themed + photo cards                                                                                    |
| Filter dimensions      | Area, Category (with sub-pills), Vibe, Date range                                                         |
| Tag taxonomy           | ~30 unique tags                                                                                           |
| Privacy posture        | No tracking · no analytics · no cookies. Itinerary save is opt-in only with explicit consent banner. Visi |
| Hosting requirement    | Any Linux/Windows server with Python; no database                                                         |
| Recurring monthly cost | \$5–\$20 typical (hosting + ~\$0.50/mo AI). See Cost Breakdown for detail.                                |
| First-year total       | ~\$80–\$150 typical. ~\$0 possible on free hosting tiers.                                                 |

## How to use this document

Pages 2–4 define *who* the platform serves — nine distinct visitor personas. Pages 5–6 list every feature, organized by audience. Pages 7–8 walk through an eight-minute demo, click by click. The remaining pages cover talking points, costs, and server migration.

# Who This Serves

Nine distinct visitor personas — each searches differently, measures success differently, and converts on different content.

A tourism platform either serves everyone weakly or serves a focused set of personas exceptionally well. Understanding who you're building for shapes navigation, content priorities, and where the chamber spends marketing budget. The nine personas below split into two macro-modes:

| Macro-mode                            | How they use the site                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Browse-first</b> (vibe-shopping)   | Visitor is shopping a feeling — "romantic weekend," "active trip," "decompression." They want vibe |
| <b>Date-first</b> (calendar-shopping) | Visitor is shopping a calendar window — "what's happening that weekend." They want fresh event     |

## The nine personas

| Persona                            | Who & trip                                            | How they search differently                                                          |
|------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>1. Romantic Weekender</b>       | Couples 35-65, Bay Area / Sac, 1-3 nights, splurge on | "Where should we have the perfect date weekend?"</i> Shopping a feeling — the        |
| <b>2. Festival Pilgrim</b>         | Any age, anchored to a specific event/dater for X.    | "What else should I do?"</i> Date-locked — searches the event                        |
| <b>3. Trail & Lake Seeker</b>      | 25-45, sporty, often gear-heavy, day/weekend nights   | "Where can I hike / bike / swim / paddle this weekend?"</i> Needs physical spe       |
| <b>4. Multi-Gen Family</b>         | Parents 30-55 with kids 5-15, sometimes grandkids     | "Will my kid parents enjoy this?"</i> Every venue gets the "will my 8-year-old       |
| <b>5. Arts & Heritage Traveler</b> | 40-70, college-educated, often retired                | "What's the depth here?"</i> Wants substance over photogenic stops. Reads            |
| <b>6. Wine Country Foodie</b>      | 30-60, food/wine enthusiasts, often wine-related      | "Where are the best things to eat?"</i> Meal-anchored trip planning. Needs prove     |
| <b>7. Wellness Refugee</b>         | 30-55, urban professional, burnt out, wife has to     | "Where can I actually be able to slow down here?"</i> Filtering for what's NOT there |
| <b>8. Local "What's Up"</b>        | Nevada County residents themselves                    | "What's going on tonight?"</i> Already knows the venues. Wants pure events           |
| <b>9. Maker Traveler</b>           | 25-65, professional or semi-retired                   | "Where do I go to find a workshop / throw a pot."</i> Shopping a SKILL, not a        |

■ **Maker Traveler highlight:** Western Nevada County is unusually well-positioned here — The Curious Forge, Wolf Craft School, ASiF, and Crafffolk Collective form a maker ecosystem few foothills counties can match. This persona has the highest revenue-per-visit (2-3 nights, workshop fee, 4-6 meals out, lodging usually unbooked from comparison sites), and the marketing story writes itself.

# How well the current site serves each persona

| Persona                  | Today      | Gap / next step                                                                                 |
|--------------------------|------------|-------------------------------------------------------------------------------------------------|
| Romantic Weekender       | ■ Strong   | Relaxed vibe, B&B sub-pill, real B&B photos already in place                                    |
| Festival Pilgrim         | ■■ Partial | Manual filtering required. Seasonal/event-anchored itineraries (planned) directly fixes this.   |
| Trail & Lake Seeker      | ■ Strong   | Hiking/Biking/Swimming/Boating/Fishing/Running tags + sub-pills shipped recently. Difficulty    |
| Multi-Gen Family         | ■■ Partial | Family vibe exists. Age-range guidance per entry isn't structured yet.                          |
| Arts & Heritage Traveler | ■ Strong   | Historic and Arts vibes both rich. Long-form storytelling is the next polish.                   |
| Wine Country Foodie      | ■ Strong   | Foodie vibe + Restaurant sub-pills + 9 wineries + 9 tasting rooms in db.                        |
| Wellness Refugee         | ■ Strong   | Recently broadened (Restorative tag, 5 new massage venues, float/spa/yoga properly captu        |
| Local "What's Up"        | ■■ Weak    | Same filtering as visitors today. A "this week" landing page + email digest would build residen |
| <b>Maker Traveler</b>    | ■■ Partial | <b>Hands-On vibe + Workshops sub-pill in place</b>, but workshop calendars (Curious For         |

## Strategic recommendation

Of the nine personas, the chamber should **over-serve four** in site content, marketing, and budget. The remaining five should be served gracefully by the platform but not be primary marketing targets.

| <b>Over-serve (focused marketing)</b><b>Why</b> |                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ■ Maker Traveler                                | Highest revenue per visit; defensible market position; The Curious Forge alone is a destination       |
| Festival Pilgrim                                | Drives concentrated peak-season tourism (Cornish Christmas, Wild & Scenic, Bluegrass).                |
| Romantic Weekender                              | Bay Area weekend market is enormous and well-suited to the foothills product. Highest per-cap         |
| Wine Country Foodie                             | Napa-alternative positioning is achievable. Sierra foothills wine region needs visibility this site c |

The other five personas (Trail & Lake, Family, Arts & Heritage, Wellness, Local) are well-served by what's built and don't require concentrated marketing — they self-discover or are served by adjacent channels (REI events, parent groups, local newspapers).

## What the platform does for each persona

A single visit to the home page should let any of the nine personas find their answer in under 30 seconds. The discovery paths are:

- **Vibe cards** (9 themed) — direct match for Romantic, Active, Foodie, Wellness, Hands-On (Maker), Family, Festivals, Arts, Historic
- **Category dropdown + sub-pills** — matches specific shopping intent (Lodging → B&Bs, Outdoor → Swimming, etc.)
- **Date filter** — Festival Pilgrim and Local "What's Up" go straight here
- **Smart Suggestions** in the itinerary modal — works for any persona once they've added 2+ items

- **Sub-pills** like ■ Workshops & Classes (Activity) or ■ Spa & Float (Wellness) — fine-grained intent

*"You don't need to serve everyone equally. You need to serve the four right personas exceptionally — and not actively annoy the other five. The platform already does the second; the marketing investment is what closes the first."*

# Feature Inventory

Everything the platform does today — built and tested.

## For Visitors

### Discovery & Filtering

| Feature             | Description                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------|
| Area filter         | Nevada City, Grass Valley, Penn Valley, North San Juan, Rough & Ready, Washington, Chicago P             |
| Category dropdown   | 15 categories — Lodging, Outdoor, Restaurant, Music venue, Wellness, etc.                                |
| Sub-pills           | 7 categories show contextual sub-types (e.g. Lodging → Hotels, B&Bs, Glamping, Campgrounds, R            |
| Themed vibes        | 9 photo cards: Historic, Arts & Music, Hands-On, Foodie, Active, Relaxed, Wellness, Family, Festiv       |
| Date filter         | Free range + quick presets: Today, This Weekend, This Week                                               |
| Auto-season hiding  | Visiting in November? Summer-only experiences disappear automatically                                    |
| Active filter chips | Each active filter shown as a chip — one-click clear                                                     |
| Tag-based search    | Six new activity tags (Hiking, Biking, Running, Swimming, Fishing, Boating) attached to every relevant e |

### Planning Tools

| Feature              | Description                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| "Add to My List"     | Heart on every card; floating itinerary button shows running count                                          |
| My Itinerary modal   | Numbered stops, removable, with hours and direct links                                                      |
| ■ Opt-in device save | <b>Privacy-by-default.</b> On first add, visitor is asked: "Save your itinerary on this device?" Yes → su   |
| ■ Share / Save link  | One tap → native phone share sheet (text, email, AirDrop, copy link). The link restores the entire itinerar |
| Cross-device handoff | Build on phone in the morning, text the link to yourself, open on iPad in the car. No login, no sync servic |
| Map view             | Leaflet/OpenStreetMap with custom gold pin markers                                                          |
| Print itinerary      | One click — formatted for paper hand-off (also "Save as PDF" from print dialog)                             |
| ■ Smart Suggestions  | Algorithmic — finds nearby experiences and matching events; shows "0.3 mi away" labels                      |
| Events inline        | Upcoming events show alongside places, filtered by the same vibe and dates                                  |
| Google Maps pin      | Custom SVG drop-pin button on each card opens directions                                                    |
| RSS feed             | /feed.rss publishes approved events for partner sites and aggregators                                       |
| Mobile responsive    | All features work on phone & tablet                                                                         |

# For the Tourism Body / Admin

## Content Management

| Feature                     | Description                                                          |
|-----------------------------|----------------------------------------------------------------------|
| Inline experience editor    | Sortable table of all 161 entries — edit, add, delete in browser     |
| Tag taxonomy editor         | Add/remove/rename tags; changes propagate to filtering immediately   |
| Drag-drop itinerary builder | Three-panel UI: itineraries · stops · browse                         |
| Events Queue                | Pending / Approved / Dismissed with filter buttons                   |
| Approve all                 | One-click bulk approval of pending events; auto-dismisses past dates |
| Auto-prune                  | Past events removed nightly; dismissed items purged after 60 days    |
| Source management           | Add/edit/disable scraper source URLs without touching code           |

## Automation

| Feature               | Description                                                                           |
|-----------------------|---------------------------------------------------------------------------------------|
| 6 scrapers            | KVMR, Eventbrite Nevada, NC Chamber, GV Chamber, The Union (RSS first), Go Nevada     |
| Auto-tagging          | Keyword pipeline — every scraped event gets baseline tags                             |
| ■ AI Categorize       | Claude Haiku refines tags, infers area & venue, scores quality (~\$0.20 per full run) |
| On-demand scraping    | "Update Now" button fires the scrape pipeline; status bar shows progress              |
| Idempotent processing | AI never re-categorizes already-processed events; cheap on subsequent runs            |
| Graceful degradation  | Site works without the server (static fallback); without an API key (no AI)           |

## Architecture & Operations

| Feature           | Description                                                                |
|-------------------|----------------------------------------------------------------------------|
| No database       | JSON files — easy backup, version control, no DBA needed                   |
| Python & Flask    | Single small server; runs on any Linux/Windows VPS                         |
| No vendor lock-in | No SaaS subscription; full code, full data ownership                       |
| Self-hostable     | Works behind Apache/Nginx; integrates with existing chamber infrastructure |
| Static-friendly   | Public site can be served as plain HTML if API isn't needed                |
| Open standards    | OpenStreetMap, RSS 2.0, plain JSON — no proprietary formats                |

# Demonstration Script

Eight-minute live walkthrough — visitor experience first, admin second.

## Setup (before the meeting)

- Run `python server.py` in the project folder
- Open `http://localhost:5000` in a fresh browser window
- Confirm **My Itinerary** is empty (refresh / clear if needed)
- Have a second tab ready on the public view, but DON'T show admin until Act 4
- Confirm the date is set in 2026 so seasonal experiences behave correctly

## Act 1 — "I'm a visitor planning a trip" · 2 min

*"Pretend you're a couple from Sacramento planning a romantic November getaway."*

| Click                                       | What to say                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Click "This Weekend" date preset            | Notice the trail and outdoor experiences that are summer-only just disappeared. The system knows it's November.          |
| Click the Festivals & Celebrations category | Overland Christmas, Victorian Christmas, and Mardi Gras appear at the top — exactly the things you'd expect in November. |
| Pause on a card                             | Point out: real photo (pulled from the property website), tags, hours, notes, the More Info link, and the heart icon.    |

## Act 2 — "Let me narrow further" · 90 sec

| Click                                    | What to say                                                                                 |
|------------------------------------------|---------------------------------------------------------------------------------------------|
| Clear date filter (X on the chip)        | Filter chips and dates clear instantly.                                                     |
| Click Relaxed vibe (alongside Festivals) | "You can layer multiple vibes. The grid recombines on the fly."                             |
| Category dropdown → Lodging              | Sub-pills appear: Hotels · B&Bs/Inns · Glamping · Campgrounds · RV Parks.                   |
| Click ■ B&Bs / Inns pill                 | "These are real Victorian B&Bs in Nevada City — Broad Street Inn, Flume's End, Piety Hill." |
| Add 2-3 favorites to the list            | The heart icon flips, the floating itinerary count increments.                              |

## Act 3 — "Building my weekend" · 2 min

| Click                                    | What to say                                                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Clear the Lodging filter                 | Back to the full grid.                                                                                 |
| Switch to Foodie vibe                    | Restaurants, wineries, breweries, and food festivals appear together.                                  |
| Add Lola Restaurant + Three Forks Bakery | Map fills, count goes up.                                                                              |
| Switch to Historic & Gold Rush vibe      | Empire Mine, Narrow Gauge Railroad Museum surface to the top.                                          |
| Add Empire Mine to the list              |                                                                                                        |
| Click the My Itinerary floating button   | Modal opens. <b>This is the magic moment.</b>                                                          |
| Scroll to ■ More to Explore              | "The system noticed everything's clustered in Grass Valley + downtown Nevada City — so it's            |
| Click + Add to List on a suggestion      | Suggestions instantly recompute — list grows, new neighbors appear.                                    |
| Click ■ View on Map                      | All stops appear on a Leaflet/OSM map with custom numbered gold pins.                                  |
| Add the first item (if fresh visit)      | <b>A small banner slides up:</b> "Save your itinerary on this device?" — explicit consent. <b>&lt;</b> |
| Close the map, click ■ Share / Save      | "Now imagine I built this on my phone — I want my partner to see it." On phone the native share        |
| Reopen the modal                         | "Notice the line at the bottom: <i>Saved on this device · Forget on this device</i> . Visitor can r    |
| Click ■ Print                            | Print preview opens — "Or save as PDF and email it. Print it for the fridge."                          |

## Act 4 — "Behind the scenes" · 90 sec

*"Now let me show you what the chamber team would actually use."*

| Click                               | What to say                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------|
| Click ■ Manage Database (top right) | Admin panel opens. Three management tabs: Experiences · Itineraries · Events Queue · Scrap     |
| Stay on the Experiences tab         | "All 161 entries — inline editable. No developer needed. Add a new venue, rename it, retag it, |
| Click ■ Events Queue                | "Where chamber staff review what's scraped overnight."                                         |
| Show pending count + filter buttons | "KVMR posted twelve new events; click Approve All; done in five seconds."                      |
| Click ■ AI Categorize               | "For about 20 cents we can have Claude clean up area names, infer venues for KVMR's vague      |
| Click ■ Scraper Sources             | "Add a new feed — say SYRCL's events RSS — paste the URL, save. The next overnight run         |

## Act 5 — Close · 60 sec

*"What this gives Nevada County:"*

- **Time saved for visitors** — they find Cornish Christmas + a B&B in 30 seconds instead of 30 minutes across five different sites



- **Local business visibility** — every chamber member appears under multiple discovery paths: vibe, area, category, sub-pill, date filter, search-like tags
- **Privacy without compromise** — opt-in itinerary save with explicit consent, cross-device share via link, no tracking, no accounts, no emails. Visitors stay anonymous; the chamber gets credit for respecting them.
- **No vendor lock-in** — open files, no database, no SaaS contract
- **Self-publishing** — the public RSS feed is something other sites and apps can pick up automatically
- **Modern AI integration** — categorization today, smart visitor suggestions next, all opt-in and bounded by cost
- **Fresh data, low maintenance** — scrapers run on a schedule; chamber staff review and approve in minutes a week

# Talking Points

Anticipated questions and the right answers.

| Question                                                 | Answer                                                                                                                  |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Can we host this ourselves?                              | Yes — single Python file, runs on any Linux box. Apache or Nginx friendly. The code can sit in a Git repo.              |
| What if a venue closes or changes hours?                 | Manage Database → click the row → edit or delete. Saved instantly. No developer. No release process.                    |
| Can we add events from [specific source]?                | Yes. Each scraper is a 100-line Python file. New sources are added by writing one matching that pattern.                |
| How fresh is the event data?                             | Last scrape timestamp shows in the admin status bar. Schedule a daily cron job for hands-off freshness.                 |
| Is the AI tagging accurate?                              | Claude Haiku gets the obvious cases right (Center for the Arts → Grass Valley). Every AI field is adjustable.           |
| What does it cost monthly?                               | Typical: \$10-\$20/month all-in (hosting + AI). Lean setup possible at ~\$1/month using a free hosting tier.            |
| Can we white-label this for the chamber?                 | Yes. Colors, logo, copy, photos all live in the HTML/CSS. No core changes needed.                                       |
| Does it work without internet?                           | The static site degrades gracefully — visitors see curated experiences but not live events. Useful as a backup.         |
| Privacy / data collection?                               | No tracking, no analytics, no cookies by default. Visitor itineraries are <b>never</b> stored without explicit consent. |
| What if a visitor wants to save / share their itinerary? | Two independent options: (1) <b>Save on this device</b> — opt-in via consent banner; survives tab closure.              |
| Accessibility?                                           | Semantic HTML, alt text on photos, keyboard navigation. WCAG-aligned but not formally audited.                          |

## Watch out for during the demo

- **Server must be running** — if you forget, the Events tab and AI button will be empty. The site degrades gracefully but the demo is much weaker.
- **Don't open the admin panel first.** Visitors first; "and here's how easy it is to maintain" second.
- **Skip the feature-list slides.** The live demo IS the pitch. Use this PDF as your prep, not your slides.
- **Suggestions need 1+ items first.** Always add something before opening the My Itinerary modal.
- **If WiFi is bad,** the Leaflet map and any uncached photos may fail. Have a fallback screenshot.
- **Audience-fit:** the Foodie / Festivals / B&Bs combination is most likely to land — they're universally appealing categories.

## Leave-behind one-liner

*"This isn't a website — it's a planning workspace for visitors and a living directory for the chamber, with AI-assisted curation built in. Privacy-respecting by default, owned by the county outright, maintained in minutes per week."*

# Cost Breakdown

What it actually costs to run, based on real measured usage.

## Recurring monthly costs

| Line item                                      | Notes                                                                                                                           | Monthly cost              |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Web hosting                                    | Any small Linux VPS (DigitalOcean, Linode, AWS Lightsail). Standard option available if no admin features needed. \$10-\$20/mo. | \$10 – \$20               |
| Domain name                                    | Optional — only if not already using existing chamber domain. Renewed annually at ~\$12-15/year.                                | \$12 – \$15               |
| SSL certificate                                | Free via Let's Encrypt; renews automatically.                                                                                   | \$0                       |
| AI event categorization                        | Anthropic Claude Haiku API. Per measured usage (~460 events) \$0.30-\$1.50 run, then \$0.01-\$0.02 per scheduled event.         | \$0.30 – \$1.50           |
| Email / user accounts                          | No user accounts in the system. No email service required.                                                                      | \$0                       |
| Database                                       | No database — JSON files. No DB hosting fees.                                                                                   | \$0                       |
| Analytics                                      | No third-party tracking by default. If desired, free options (Plausible self-hosted, GoatCounter) available.                    | \$0                       |
| <b>Realistic monthly total</b>                 |                                                                                                                                 | <b>\$5 – \$25</b>         |
| Bare-minimum monthly total (free hosting tier) |                                                                                                                                 | \$0.30 – \$1.00 (AI only) |
| Typical chamber operation (paid VPS + AI)      |                                                                                                                                 | \$10 – \$20               |

## One-time / annual costs

| Line item                  | Notes                                                                                                         | Cost        |
|----------------------------|---------------------------------------------------------------------------------------------------------------|-------------|
| Initial setup & deployment | Configure server, install Python, deploy code, set up daily scraper, one-time point domain. One person-hours. | one-time    |
| Domain renewal             | If purchasing new domain (e.g. visitwesternnevada.org).                                                       | ~\$12-15/yr |
| Content review (annual)    | Walking through all 161 experiences once a year to verify hours, staff, etc. Roughly 4-6 staff hours.         | \$200s/yr   |

## AI cost detail (the only variable expense)

The AI categorization cost is the only line item that scales with usage. Here's exactly what it costs based on measured token counts:

| Run pattern                                         | Per-run cost | Monthly equivalent |
|-----------------------------------------------------|--------------|--------------------|
| First-time bulk categorization (current 460 events) | \$0.29       | one-time           |

| Run pattern                                               | Per-run cost            | Monthly equivalent           |
|-----------------------------------------------------------|-------------------------|------------------------------|
| Daily scrape + categorize (~10-30 new events/day)         | \$0.01 - \$0.02 per run | \$0.30 – \$0.60 / mo         |
| Weekly scrape + categorize (~70-200 new events/week)      | \$0.04 - \$0.13 per run | \$0.16 – \$0.52 / mo         |
| Quarterly full re-categorize (with --force)               | \$0.30 per run          | \$0.10 / mo equiv.           |
| Realistic chamber operation (weekly + occasional re-runs) |                         | <b>~\$0.50 – \$1.00 / mo</b> |

## Built-in cost protection

- **Hard usage cap on Anthropic side** — recommended \$5/month spending limit set in API console. Cannot ever exceed it; emails sent at 50% / 75% / 100%.
- **Idempotent processing** — events that have already been categorized are skipped automatically. No accidental re-billing on the same data.
- **Optional, not required** — the entire AI feature can be turned off (no API key set) and the site still functions on raw scraper data.
- **Quality filter blocks waste** — events flagged "low quality" by AI are auto-hidden from the public site, reducing noise without manual review.

# Annual Cost Summary

Three realistic scenarios for the first year of operation.

## Three operating scenarios

| Scenario          | Lean                          | Typical                  | Premium                        |
|-------------------|-------------------------------|--------------------------|--------------------------------|
| Hosting           | Free tier (e.g. Oracle Cloud) | \$5/mo VPS               | \$20/mo managed VPS            |
| Domain            | Use existing chamber domain   | New domain (~\$15/yr)    | Premium domain                 |
| AI categorization | Disabled or rare              | Weekly runs (~\$0.50/mo) | Daily runs + re-runs (~\$1/mo) |
| Content review    | 4 hrs / year staff time       | 6 hrs / year             | Quarterly review               |
| Backups           | Manual quarterly              | Automated weekly         | Automated daily off-site       |
| First-year total  | ~\$0 – \$50                   | ~\$80 – \$150            | ~\$300 – \$500                 |
| Subsequent annual | ~\$0 – \$20                   | ~\$70 – \$130            | ~\$280 – \$480                 |

## What this means in practice

For comparison, a commercial off-the-shelf tourism platform (CrowdRiff, Simpleview Destination Stack, or similar) typically runs **\$5,000 – \$30,000 annually** with multi-year contracts and per-feature pricing. The Nevada County Experience build delivers comparable visitor-facing capability for one to three orders of magnitude less, and the chamber owns all the code and data.

## What's NOT a cost

- **Per-listing fees** — every chamber member can be listed at no incremental cost.
- **Per-event fees** — scraped events come in for free; AI categorization is the only variable cost.
- **Per-visitor fees** — no rate-based pricing; if traffic 10×'s tomorrow, hosting cost barely changes.
- **License fees** — no proprietary software. Python, Flask, Leaflet, OpenStreetMap, all open source.
- **Vendor termination fees** — owned outright. No vendor to leave.
- **Migration costs** — JSON files are portable; can switch hosts in an afternoon.
- **Privacy compliance overhead** — no analytics, cookies, or trackers means no GDPR/CCPA banners, no privacy review, no consent platform fees. Itinerary save is opt-in by design.

## Recommended starting budget

**\$200 in the first year** covers everything: domain (\$15), modest VPS hosting (\$60/year), AI categorization (\$10), and a small reserve. Subsequent years drop to ~\$85 if domain doesn't need renewal that year. No staff training cost — the admin panel is intentionally simple enough that anyone who can edit a spreadsheet can maintain it.

# Migration to a County Server

Step-by-step deployment from local development to production hosting.

This is the full sequence to move the platform from a developer laptop to a production server the chamber controls. Estimated effort: **one focused half-day** for someone comfortable with Linux, **two days** for someone learning as they go. No code changes are required — the same files run locally and in production.

## Phase 0 — Decisions before you start

Lock these down first. Each affects what comes next.

| Decision             | Recommended                                | Alternative                                            |
|----------------------|--------------------------------------------|--------------------------------------------------------|
| Hosting provider     | DigitalOcean (\$6/mo droplet) or Linode    | AWS Lightsail · chamber's existing host · county-IT VM |
| Operating system     | Ubuntu 24.04 LTS                           | Debian 12 · RHEL 9 · Windows Server (more work)        |
| Domain               | New: visitwesternnevada.org (~\$15/yr)     | Subdomain: experience.nevadacountychamber.com          |
| Server account owner | Chamber / county directly                  | Contracted dev (lock-out risk)                         |
| SSL certificate      | Free Let's Encrypt (auto-renew)            | Paid cert (no real benefit)                            |
| Backups              | Provider snapshots + monthly off-site copy | Snapshots only · weekly tar to chamber NAS             |

**Decision-point:** if the county already has a web server (Apache or nginx running on a county-IT box), you can run this *next to* the existing site. If not, a fresh small VPS is cleaner and the cost is trivial.

## Phase 1 — Pre-flight checklist

Confirm all of these before continuing:

- Domain registered (or subdomain delegated)
- DNS access confirmed — you can edit A records
- Anthropic API key created at console.anthropic.com (with \$5/mo cap set)
- SSH key generated locally (id\_ed25519 or id\_rsa)
- You can SSH and copy files between your laptop and the server
- Cloudflare or similar set up (optional — adds free DDoS + caching)

# Phase 2 — Provision the server

## 2a. Buy the droplet

DigitalOcean → Create → Droplet → **Ubuntu 24.04 LTS** → Basic plan, \$6/mo (1 GB RAM, 25 GB SSD) → datacenter near the West Coast (San Francisco 3) → add your SSH key → name it nevada-county-experience → Create.

## 2b. First login + lockdown

```
# From your laptop
ssh root@<droplet-ip>

# Create a non-root user, give them sudo
adduser ncexp
usermod -aG sudo ncexp
mkdir -p /home/ncexp/.ssh
cp ~/.ssh/authorized_keys /home/ncexp/.ssh/
chown -R ncexp:ncexp /home/ncexp/.ssh
chmod 700 /home/ncexp/.ssh
chmod 600 /home/ncexp/.ssh/authorized_keys

# Disable root SSH login
sed -i 's/PermitRootLogin yes/PermitRootLogin no/' /etc/ssh/sshd_config
sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config
systemctl restart ssh
exit

# From now on, SSH as ncexp
ssh ncexp@<droplet-ip>
```

## 2c. System packages

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3 python3-pip python3-venv \
    nginx git ufw certbot python3-certbot-nginx \
    chromium-browser chromium-chromedriver

# Firewall: SSH + HTTPS only
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
```

# Phase 3 — Deploy the application

## 3a. Get the code onto the server

**Recommended:** push to a private GitHub repo from your laptop, clone on the server. **Alternative:** zip the folder and use scp.

```
# On your laptop (one-time)
cd "C:\Users\Curious Forge\Documents\nevada-county-experience"
git init
git add .
git commit -m "Initial commit"
gh repo create nevada-county-experience --private --source=. --push

# On the server
cd /home/ncexp
git clone https://github.com/<your-user>/nevada-county-experience.git
cd nevada-county-experience
```

## 3b. Python environment

```
cd /home/ncexp/nevada-county-experience
python3 -m venv .venv
source .venv/bin/activate
pip install flask requests beautifulsoup4 lxml selenium \
        webdriver-manager python-dateutil reportlab
deactivate
```

## 3c. Create the secrets file

scraper/config.py is gitignored, so the keys never leave this server.

```
cat > scraper/config.py <<'EOF'
GOOGLE_PLACES_API_KEY = "<paste from your local config.py>"
ANTHROPIC_API_KEY = "sk-ant-api03-..."
CENTER_LAT = 39.2350
CENTER_LNG = -121.038
RADIUS_M = 35000
MIN_RATINGS = 5
MIN_RATING = 3.5
EOF
chmod 600 scraper/config.py
```

## 3d. Smoke test

```
source .venv/bin/activate
python server.py
# Should show "Running on http://127.0.0.1:5000"
# Press Ctrl+C
deactivate
```



## Phase 4 — Run as a persistent service

systemd auto-starts the server on reboot and auto-restarts if it crashes. Logs go to journalctl.

```
sudo tee /etc/systemd/system/ncexp.service > /dev/null <<'EOF'
[Unit]
Description=Nevada County Experience web app
After=network.target

[Service]
Type=simple
User=ncexp
WorkingDirectory=/home/ncexp/nevada-county-experience
Environment="PATH=/home/ncexp/nevada-county-experience/.venv/bin"
ExecStart=/home/ncexp/nevada-county-experience/.venv/bin/python \
    server.py --host 127.0.0.1 --port 5000
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable ncexp
sudo systemctl start ncexp
sudo systemctl status ncexp      # should show "active (running)"

# View logs anytime:
journalctl -u ncexp -f
```

## Phase 5 — Reverse proxy + HTTPS

Flask shouldn't face the internet directly. nginx sits in front and terminates HTTPS via Let's Encrypt.

### 5a. Point DNS to the server

```
# At your registrar:
A    visitwesternnevada.org    -> <droplet-ip>
A    www.visitwesternnevada.org -> <droplet-ip>

# Wait 5-30 min, then verify from your laptop:
dig visitwesternnevada.org
```

### 5b. nginx config

```
sudo tee /etc/nginx/sites-available/ncexp > /dev/null <<'EOF'
server {
    listen 80;
    server_name visitwesternnevada.org www.visitwesternnevada.org;

    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    # Cache static assets aggressively
```

```
location ~* \.(jpg|jpeg|png|webp|gif|svg|css|js|woff2|ico)$ {
    proxy_pass http://127.0.0.1:5000;
    expires 7d;
    add_header Cache-Control "public, immutable";
}
}
EOF
```

```
sudo ln -s /etc/nginx/sites-available/ncexp /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
```

## 5c. Add HTTPS

```
sudo certbot --nginx -d visitwesternnevada.org -d www.visitwesternnevada.org
# Enter email, agree to TOS, choose "redirect HTTP to HTTPS"
```

Certbot adds a cron job that auto-renews certificates 30 days before expiry. Set and forget.

## Phase 6 — Schedule the scraper

Nightly scrape at 3 AM, AI categorization 30 minutes later. Chamber staff review pending events the next morning.

```
# Edit the user's crontab
crontab -e

# Add these two lines:
0 3 * * * cd /home/ncexp/nevada-county-experience && \
    .venv/bin/python scraper/event_scraper.py >> /var/log/ncexp-scrape.log 2>&1
30 3 * * * cd /home/ncexp/nevada-county-experience && \
    .venv/bin/python scraper/ai_categorize.py >> /var/log/ncexp-ai.log 2>&1

# Make the log files writable
sudo touch /var/log/ncexp-scrape.log /var/log/ncexp-ai.log
sudo chown ncexp:ncexp /var/log/ncexp-scrape.log /var/log/ncexp-ai.log

# Run the scraper once manually to verify Selenium + Chromium work:
cd /home/ncexp/nevada-county-experience
.venv/bin/python scraper/event_scraper.py --site "Nevada City Chamber"
```

## Phase 7 — Backups

Two layers: provider snapshots for full-system recovery, plus a daily JSON dump for granular event/itinerary recovery.

### 7a. Provider snapshots (\$1/mo on DigitalOcean)

DO Console → Droplet → Backups → Enable. Weekly automatic, 4-week retention.

### 7b. Daily JSON backup

```
mkdir -p /home/ncexp/backups
cat > /home/ncexp/backup.sh <<'EOF'
#!/bin/bash
DATE=$(date +%Y%m%d)
cd /home/ncexp/nevada-county-experience
tar czf /home/ncexp/backups/ncexp-$DATE.tar.gz \
    scraper_output/events.json scraper/sources.json scraper/config.py
# Keep only last 30 days
find /home/ncexp/backups/ -name 'ncexp-*.tar.gz' -mtime +30 -delete
EOF
chmod +x /home/ncexp/backup.sh

# Schedule daily at 4 AM (after scrape + AI complete)
crontab -e
# Add:
0 4 * * * /home/ncexp/backup.sh
```

### 7c. Off-site copy (optional but recommended)

Once a month, copy the latest backup tarball to chamber NAS, Google Drive, or another cloud. Even a manual quarterly download protects against catastrophic provider loss.

## Phase 8 — Test & cutover

| Test                             | How to verify                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| Public site loads at https://... | Browse to the domain. Confirm padlock + "https" + green/normal cert.                              |
| All vibes show experiences       | Click each of the 9 vibe cards; each should return matches.                                       |
| Events tab populates             | Should show ~460 approved events. If empty, server isn't reaching events.json.                    |
| Date filter works                | Pick "This Weekend" — events filter; some experiences hide based on season.                       |
| Mobile share works               | On a phone, build a list, hit Share. Native share sheet should appear.                            |
| Admin panel accessible           | Click ■ Manage Database. All four tabs should render.                                             |
| Scraper runs by hand             | <code>sudo -u ncexp .venv/bin/python scraper/event_scraper.py --site "Nevada City Chamber"</code> |
| AI Categorize button works       | Manage Database → Events Queue → ■ AI Categorize → enter 3 → confirm enrichment appears in ev     |
| Cron jobs scheduled              | <code>sudo crontab -u ncexp -l</code> should show the 3 AM and 3:30 AM lines.                     |
| Backups running                  | Wait until tomorrow morning, confirm <code>/home/ncexp/backups</code> has a new file.             |
| Service restarts                 | <code>sudo systemctl restart ncexp</code> ; site comes back within 2 seconds.                     |
| HTTPS auto-renews                | <code>sudo certbot renew --dry-run</code> should report success on all certs.                     |

## Phase 9 — Handoff to chamber

After the technical setup, hand the chamber team a small kit so they can operate the system day-to-day.

- **Admin URL bookmarked** on the chamber's computers — they only need to remember the public URL; the admin panel is reached from a button on the home page
- **Anthropic billing alerts** set up to email the chamber's ops manager at 50% / 75% / 100% of monthly cap
- **One-page cheat sheet** covering: how to approve events, how to add a venue, how to revoke an event after approval, how to trigger a manual scrape
- **Emergency contact** identified — who fixes it if the server is down? County IT? A contractor on retainer? The original developer?
- **Annual review on the calendar** — once a year, walk through all 161 experiences, verify hours/URLs/photos, prune anything closed
- **Backup test** — once before going live, restore from a backup to a test droplet to confirm the recovery process works
- **Domain renewal calendar reminder** — set 90 days before expiry so it never lapses

### Estimated effort & timeline

| Path                  | Time to live               | Approximate cost  |
|-----------------------|----------------------------|-------------------|
| Confident Linux admin | 4 hours (single afternoon) | ~\$400 contracted |

| Path                       | Time to live                | Approximate cost |
|----------------------------|-----------------------------|------------------|
| Self-taught learner        | 2 days (with research time) | time only        |
| County IT department       | Schedule-dependent          | internal         |
| Hosted-by-developer option | Same day                    | ~\$50/mo managed |

## Common pitfalls to watch for

- **Selenium / ChromeDriver version mismatch** — if scrapes fail with "session not created", apt usually keeps chromium-browser and chromium-chromedriver in sync, but a manual chromedriver download may be needed
- **Firewall too tight** — make sure ports 80 and 443 are open via ufw before running certbot, or HTTPS setup will fail
- **Time zone** — server defaults to UTC; cron jobs run in UTC. Either set the server to America/Los\_Angeles or schedule jobs in UTC (3 AM PT = 11 AM UTC)
- **File permissions on scraper output** — events.json must be writable by the ncexp user, or the scraper silently fails
- **Anthropic API key in repo** — confirm scraper/config.py is gitignored before first git push